

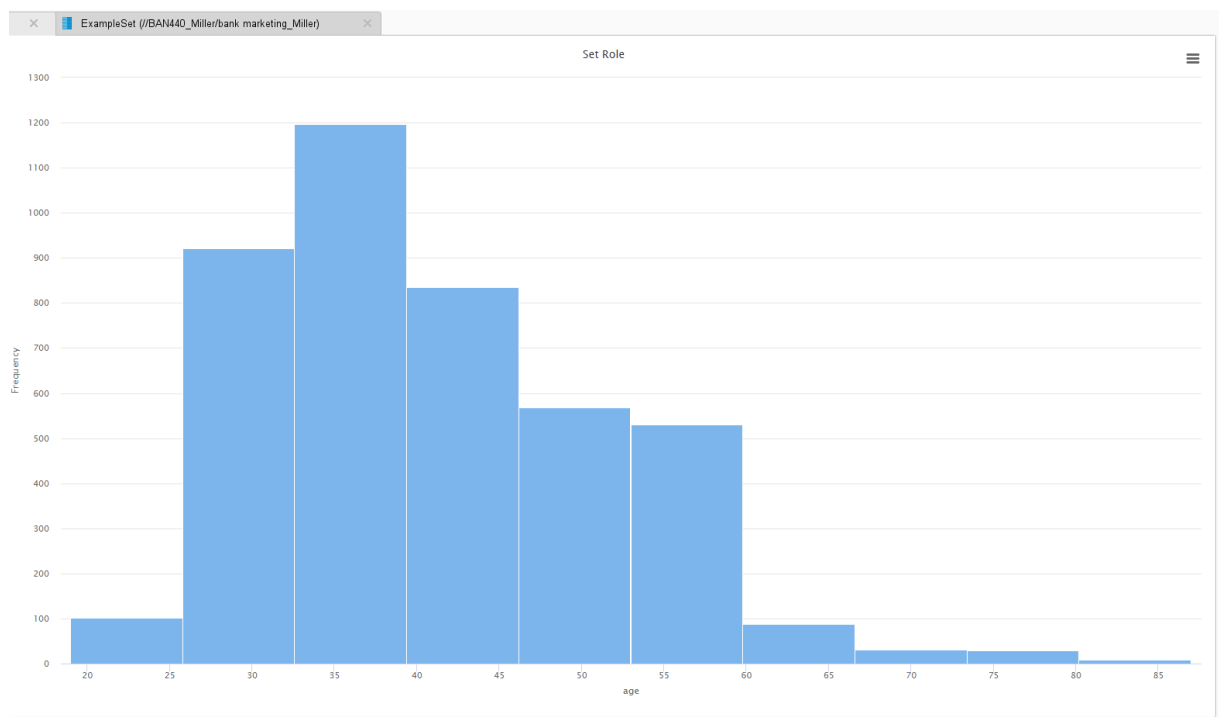
Bank Marketing Final Project

Exploratory Data Analysis

Overview of Dataset: This dataset of a bank's observations of their marketing campaigns, and how their clients interact with them contains 4311 records with 17 variables, 16 being input and 1 being a target variable. This dataset is heavily skewed, as the large majority (88.5% of people) are not subscribed, with only 11.5% being subscribed.

Descriptive Statistics of Age and Job Type

Age(numeric) represents the bank's clients age, as shown in the table below, the youngest client is 19, the oldest is 87, and the average age is about 41. The histogram also shows us that this distribution is right skewed, with the number of people contacted by their marketing campaign being highest around 30-40, and a gradual fall off from there

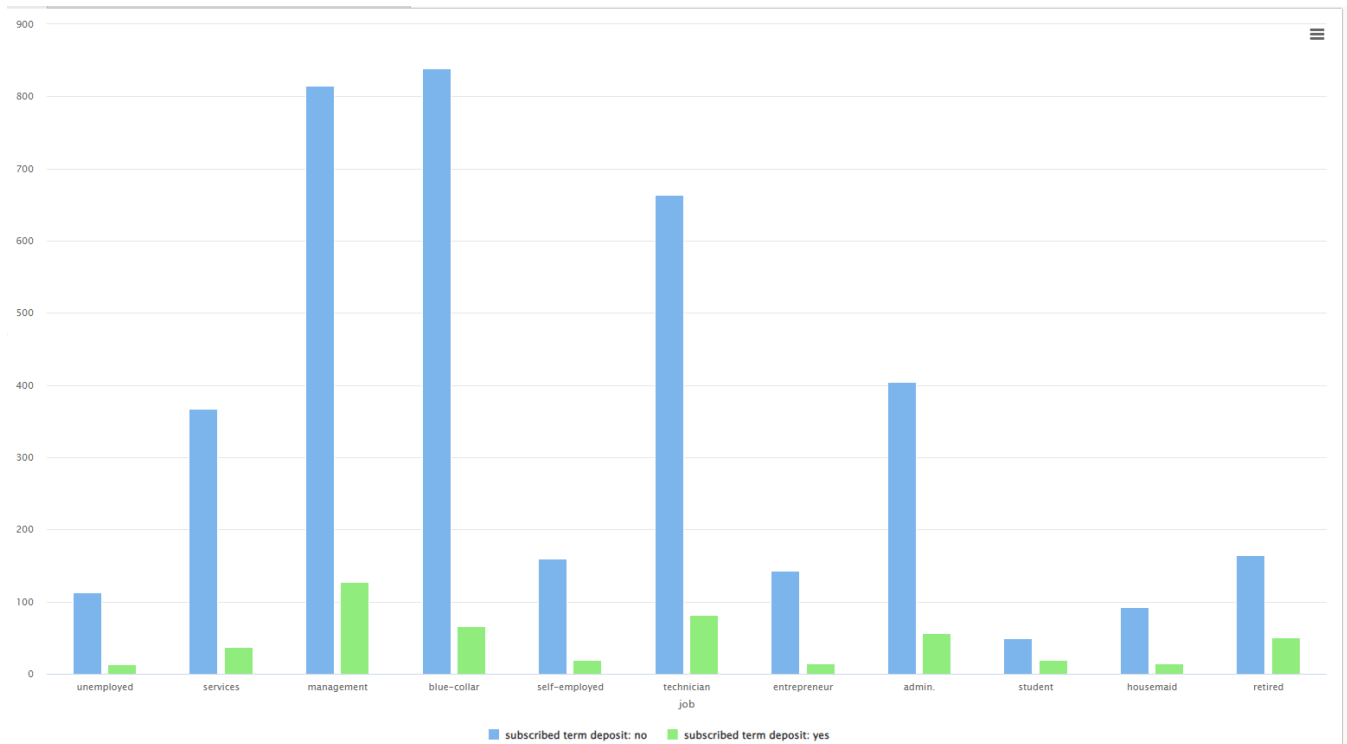


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Job type(categorical) is a variable that shows the 11 most popular jobs that the clients of the bank work at, the two most popular jobs are management and blue collar far and away, at over 900 clients in each field, with technician jobs being close 3rd at 750 clients. Students and maids are the least represented jobs in this dataset. The bar chart below shows us that even though there are the most blue collar and management jobs, their ratios of subscribed to not is much less significant than the lesser categories, like students. Who have proportionally higher rate of subscriptions.



Preprocessing

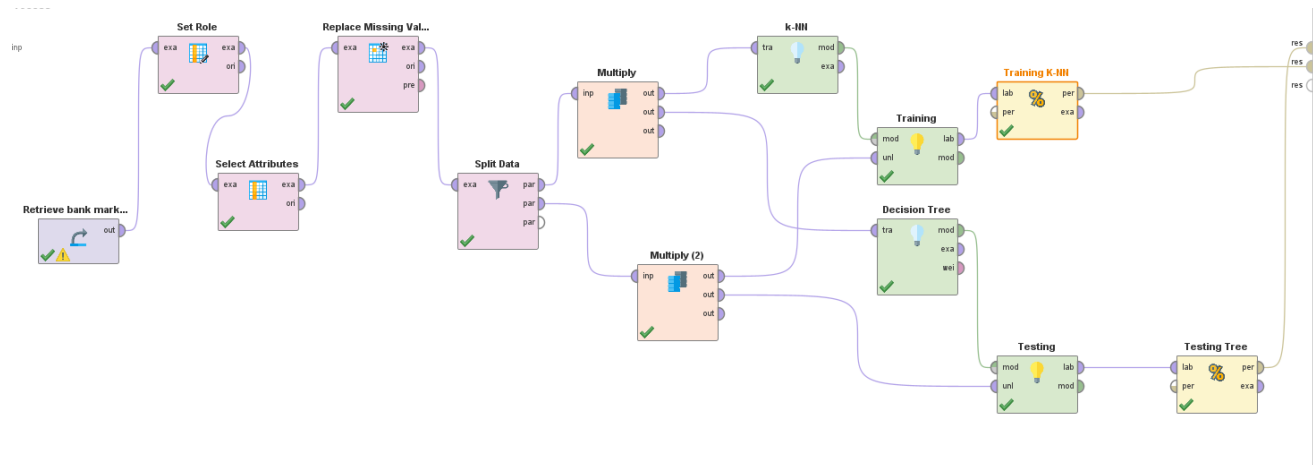
Of the 16 variables, the ones I chose were age, job, marital status, education level, and current balance, as well as the subscribed term deposit. For the missing values I used a remove missing values operator. The data was split 70/30 between the larger training and smaller testing

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sets. The below table shows the model created in RapidMiner to attain this data. For the two models needed, I used a K-NN model, and a Decision Tree.



Evaluation and comparison of the Models.

Decision tree.

PerformanceVector (Training K-NN)

PerformanceVector (Testing Tree)

PerformanceVector

PerformanceVector:

accuracy: 88.48%

ConfusionMatrix:

True:	no	yes
no:	1144	149
yes:	0	0

K-NN model

PerformanceVector

```
PerformanceVector:
accuracy: 88.01%
ConfusionMatrix:
True:   no    yes
no:     1134   145
yes:     10    4
```

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Comparison

Both models performed relatively the same in terms of accuracy, with only a 0.47% difference between the two models in favor of the K-NN model. However, the confusion matrices showed which model was truly better, with the Decision tree accurately predicting 4 subscribers, and predicted 14 subscribers total, while the K-NN model predicted no for every client, which makes sense due to the large imbalance of the data, this allowed the K-NN model to achieve a higher accuracy, but it doesn't really predict anything successfully. This shows that the Decision tree is the better model for this dataset despite the lower accuracy, it is much more valuable of a business intelligence tool than the other model.

Business Interpretation

This bank could use a decision tree to better understand and predict which clients to invest more effort into marketing towards, which would create better conversion rates, less effort, and less resources being put into clients that are most likely not to subscribe. This would be a much more cost-effective marketing strategy.